ROBIN SAINI

Lead / Senior Data Engineer | Snowflake | Python & SQL | Spark | Airflow & dbt | Agentic AI Data Engineering

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Right to work in the UK (ILR) | Eligible for SC clearance | Notice period: 12 weeks

PROFESSIONAL SUMMARY

Lead / Senior Data Engineer with 15+ years in software and data engineering (10+ in data engineering), all in regulated UK banking. Currently lead data engineering for all agentic AI products in NatWest's enterprise innovation team, taking them from proof of concept to production: over 90% less manual case-handling time and quality checks on 300+ commercial onboarding applications a day. Previously led the team that built a 50+ source, 1TB+/month Snowflake ESG and climate data platform. Hands-on every day in Python, SQL, PySpark, Airflow and dbt, with CI/CD, automated testing and data-quality controls built into delivery. Open to permanent lead roles and contract engagements.

KEY SKILLS

- **Data Engineering:** Python (advanced), SQL (expert), PySpark / Apache Spark (incl. AWS EMR Serverless), Apache Airflow (AWS MWAA), dbt, StreamSets, batch ELT/ETL at scale
- **Data Platforms:** Snowflake (architecture, clustering, materialised views, performance and cost tuning), PostgreSQL, Teradata, IBM DB2, data lake design, dimensional and analytical modelling
- **AI Data Engineering:** MCP servers and agent tool integration, RAG / retrieval and context pipelines, OpenAI / Agents SDK, LangGraph, Pydantic data contracts, LLM-as-a-judge evaluation, AI observability; AI-assisted engineering (Kiro CLI, Claude Code)
- **Cloud & DevOps:** AWS (S3, Glue, EMR, Athena, SageMaker), GitLab CI/CD, Docker, Git, Splunk; Terraform (maintaining configuration), GCP
- **Quality & Leadership:** data quality gates, audit trails, GDPR-aligned handling, automated testing, incident management; team leadership (up to 5 engineers), mentoring, engineering standards, stakeholders, Agile

SELECTED IMPACT

- **>90% reduction** in manual case-handling time through agentic decision-support workflows; AI-assisted due-diligence checks on **300+ applications a day**.
- **50+ sources, 1TB+ a month** on the Snowflake ESG and climate data platform, delivered as team lead.
- **40% faster** batches, **30% fewer** data quality incidents (NatWest); **50% lower** ETL load times, **45% fewer** data errors (Lloyds).

PROFESSIONAL EXPERIENCE

NatWest Group, London (Hybrid) | Senior Data Engineer - Agentic Systems & LLM Deployment | Aug 2025 - Present

- Lead the data engineering function for all agentic AI products in the enterprise innovation team, from proof of concept to production, including a complaints decision-assist product, an LLM-as-a-judge evaluation service and a contact-centre assist programme.
- Built the data engineering behind decision-support workflows that cut manual case-handling time by over 90%, and an AI-assisted due-diligence capability supporting quality checks on 300+ commercial onboarding applications a day.
- Designed a reusable, PII-guardrailed MCP server (bounded JSON slices, role-based field allow-lists, full audit trail) that multiple agents consume with no data-access code, standardising access to core banking, CRM and case-management systems.
- Built RAG / retrieval and context pipelines; debugged retrieval quality at source (segment filtering, candidate-pool sizing, confidence thresholding), materially improving relevant-result recall. Cut agent data-hydration latency with warmed persistent sessions and response caching.

- Built an LLM-as-a-judge evaluation pipeline (with a batch REST path) scoring model decisions into Snowflake, and PySpark pipelines on AWS EMR Serverless for model monitoring (accuracy, precision, recall, F1, LLM token usage and latency), orchestrated in Airflow.
- Introduced Snowflake CI/CD and automated testing for data loaders; built a resilient Snowflake writer with jittered back-off, reducing deployment issues to near zero.
- Designed data contracts, Pydantic validation, quality gates and audit trails to meet banking and financial-crime compliance standards; built Splunk observability for agent execution and decision quality. Deliver via GitLab CI/CD and Docker.

NatWest Group, Edinburgh | Senior Data Engineer - Climate Analytics (ESG Data Platform) | Jul 2023 - Aug 2025

- Led a team of Data Engineers delivering NatWest's ESG and climate data platform on Snowflake: 50+ internal and external sources, 1TB+ a month, for analytics and regulatory reporting. Layered dbt project (RAW, INT, PRS, presentation) orchestrated by Airflow on AWS MWAA.
- Cut batch processing times by 40% through PySpark and dbt optimisation.
- Reduced data quality incidents by 30% with metadata-driven ingestion (new sources by config, not code), idempotent re-run guards, dbt snapshots and automated dbt tests; cut Snowflake compute cost with per-model warehouse routing and transient tables.
- Delivered production climate risk models with Data Architects and Data Scientists; defined engineering standards and data product frameworks; mentored junior engineers.

NatWest Group, Edinburgh | Senior Data Engineer - Retail Analytics | Sep 2021 - Jul 2023

- Engineered enterprise batch pipelines in Python, SQL, Snowflake, PySpark and StreamSets for the Retail Data & Analytics Decisioning team, including PySpark pipelines for Customer Lifetime Value analytics.
- Designed Snowflake data models for high-volume analytical workloads, tuned with clustering, partitioning and materialised views.

Tata Consultancy Services (client: Lloyds Banking Group), Manchester & Gurugram | Data Engineer - Enterprise Data Platform | Jun 2019 - Sep 2021

- Led a 5-member Data Services team running the Lloyds enterprise data platform: Teradata warehouse, reporting applications and data support for analytics teams.
- Designed and delivered an ML-ready data lake on AWS for data science programmes; supported the IBM DB2 operational data store; ran a Google BigQuery proof of concept for Teradata modernisation.
- Tuned SQL and batch applications to cut CPU consumption; led Teradata upgrades and disaster recovery; gave 24/7 support for business-critical, regulatory and legal applications and resolved a critical production incident in under 12 hours.

Tata Consultancy Services (client: Lloyds Banking Group) | Enterprise Data Warehouse Engineer | Feb 2016 - May 2019

- Built ETL pipelines that cut data load times by 50%; implemented a GDPR-compliant data governance framework that reduced data errors by 45%.

UST Global, Gurugram | Senior Software Engineer (Mainframe) | Apr 2015 - Feb 2016

- Built and supported COBOL / DB2 mainframe applications under ITIL incident, problem and change management; contributed to high-level designs with business analysts and mentored junior team members.

Sopra Steria, Noida | Software Engineer (Mainframe) | Apr 2011 - Mar 2015

- Trained in mainframe technology, then built and supported COBOL / DB2 systems for a UK retailer; prepared detailed designs from business requirements under ITIL processes.

EDUCATION & CERTIFICATIONS

- B.Tech, Computer Science, Gautam Buddh Technical University, India (2006 - 2010)
- Big Data on AWS, Amazon Web Services (May 2022) | Data Engineering Nanodegree, Udacity (May 2022) | Google Cloud Engineering Specialization, Coursera